

Cairo University

Faculty of Computers and Artificial Intelligence



CS251

**Intro. to Software
Engineering**

Personal Budgeting Software

Software Requirements Specifications

Version 1.0

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CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

Contents

Team.....	3
Document Purpose and Audience.....	3
Introduction	3
Software Purpose.....	3
Software Scope.....	4
Definitions, acronyms, and abbreviations	4
Requirements.....	5
Functional Requirements.....	6
Non Functional Requirements	8
System Models	10
Use Case Model.....	10
Enriched User Stories	13
System Navigation Map	41
Tools	41
Ownership Report.....	42



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

Team

ID	Name	Email
20240613	Manal Abdelazem Abdelrahman	20240613@stud.fci-cu.edu.eg
20240207	Renad Tharwat Yehia	20240207@stud.fci-cu.edu.eg
20240637	Noura Mohamed Mokhtar	20240637@stud.fci-cu.edu.eg
20242355	Menna Tarek saad	20242355@stud.fci-cu.edu.eg

Document Purpose and Audience

Document purpose: the main purpose of this document is to clearly define the functional and non-functional requirements of the system. The document provides a detailed description of the system's features and functionalities. It helps ensure that all stakeholders have a clear understanding of the system before development begins. The SRS also ensures that the final system meets the needs of its users by clearly documenting the expected behavior and system capabilities.

Target Audience: it is intended for several groups involved in the development and use of the system, including:

- 1-Project Manager: to plan, organize, and monitor the project activities based on the defined requirements.

- 2- Software Developers: to understand the required system functions and implement them correctly.

- 3- Client / Product Owner: to ensure that the system addresses the intended needs and includes the required features.

- 4- End Users: to understand the main capabilities of the system and how it is intended to support personal financial management

Introduction

Software Purpose

The purpose of this system is to help individuals manage their personal finances efficiently by providing a simple and user-friendly platform for tracking income, expenses, and savings. Many people face difficulties in controlling their spending and understanding where their money goes.



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

Therefore, this system aims to provide clear insights into users' financial activities and help them make better financial decisions. The system is designed to support users in organizing their financial life, reducing unnecessary expenses, and achieving their financial goals such as saving money for travel, education, or important purchases. By using this system, users will be able to develop better financial habits and improve their overall financial stability.

Software Scope

The proposed system is a personal budgeting application that allows users to record their financial transactions, including income and expenses. Users will be able to categorize their expenses (such as food, transportation, bills, and entertainment), create budgets for different categories, and monitor their spending over time. The system will also provide features such as setting financial goals, tracking progress, and generating simple reports and visual insights (charts and summaries) to help users better understand their financial behavior. The application is intended to be available as a mobile and/or web application to ensure accessibility and ease of use. The system focuses on simplicity and usability, especially for users in Egypt and the Arab region who may not currently use advanced financial management tools.

Definitions, acronyms, and abbreviations

- Definitions & Abbreviations:

Phase	Definition
Budget	A plan that helps users manage their income and expenses over a specific period of time.
Income	Money received by the user from sources such as salary, business, or gifts.
Expense	Money spent by the user on daily needs such as food, bills, and transportation.
Category	A way to group expenses (e.g., food, transport, entertainment).



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

Transaction	Any financial activity recorded in the system (income or expense).
Financial goal	A target set by the user to save money for a specific purpose.
Dashboard	The main screen that shows a summary of the user's financial data.
Report	A summary (charts or text) that shows spending behavior and trends.
E-wallet	A digital tool used to store money and make electronic transactions.
API	Application Programming Interface used to connect the system with external services.
UI	User Interface, the part of the system that users interact with.
<u>UX</u>	User Experience, how users feel when using the system.
DB	Database where all user data is stored.
OTP	One-Time Password used for secure authentication.
PIN	Personal Identification Number used for user verification.

Requirements



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Project: Personal Budgeting Software

Software Requirements Specifications

Functional Requirements

1- User Registration and Login:

R1.1 Requirement: The system shall allow new users to create a new account by providing basic information such as name, email, and password.

R1.2 The system must allow registered users to log in using their email and password.

R1.3 The system must display an error message if the login credentials are incorrect.

R1.4 The system must validate the user's login credentials before granting access to the system.

R1.5 After successful login, users are redirected to their personal dashboard or home page.

2- Add Income:

R2.1 Requirement: The system must allow users to add a new income record, enter the income amount, specify the income source and date of the income.

R2.2 The system must store the income information in the system database.

R2.3 The system must update the user's total balance after adding each income.

3- Add Expense:

R3.1 Requirement: The system must allow users to add a new expense record, enter the expense amount, select an expense category, add optional notes for the expense.

R3.2 The system must store the expense information in the system database.

R3.3 The system must update the expense summary automatically.

4- Manage Expense Categories:

R4.1 Requirement: The system must provide predefined expense categories.

R4.2 The system must allow users to select a category when adding an expense, edit existing categories or delete custom categories.



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Project: Personal Budgeting Software

Software Requirements Specifications

R4.3 The system allows users to create new custom categories if they not exist.

5- Create Monthly Budget:

R5.1 Requirement: The system allows users to create a monthly budget.

R5.2 The system must allow users to define spending limits for different categories and store the budget information.

R5.3 The system must compare user expenses with the defined budget limits.

R5.4 The budget status should update automatically when new expenses are added

6- Track Budget Usage:

R6.1 Requirement: The system must calculate the remaining budget for each category.

R6.2 The system must display budget usage information to the user and update the remaining budget after each recorded expense.

R6.3 The system must alert the user if the budget limit is close to being exceeded.

7- Generate Financial Reports:

R7.1 Requirement: The system must display summaries of income and expenses.

R7.2 The system must generate monthly financial reports for users.

R7.3 The system allows present financial data using charts or graphs.

8- Set Savings Goals:

R8.1 Requirement: users can create saving goals, specify the target savings goal.

R8.2 The system tracks the progress toward the savings goal and in the end display the percentage of goal completion.

9- View Transaction History:



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Project: Personal Budgeting Software

Software Requirements Specifications

R9.1 Requirement: The system must store all financial transactions and allow users to view their all transactions.

R9.2 The system must allow users to filter transactions by date or category.

10- Dashboard Overview:

R10.1 Requirement: The system must display a dashboard summarizing the user's financial information.

R10.2 The dashboard must display the total income, total expense, remaining balance.

R10.3 The dashboard must display recent financial transactions.

Non Functional Requirements

Performance	• The login screen shall load within 3 seconds. • Adding a new transaction (income/expense) shall take less than 2 seconds. • Generating monthly reports and charts shall take less than 5 seconds. • The dashboard with spending overview shall load within 4 seconds.
Scalability	• The system shall support up to 10,000 simultaneous users without performance degradation. • The database shall handle up to 1 million transactions per user without slowing down.



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

Availability	The system shall be available 99.5% of the time (except scheduled maintenance). • Maximum downtime shall not exceed 4 hours per month.
Security	All user financial data shall be encrypted using AES-256 encryption (at rest). • All data transmitted between the app and server shall use TLS 1.3 protocol. • User passwords shall be hashed using bcrypt before storing in database. • The system shall implement OAuth 2.0 for bank account integration. • Multi-factor authentication (MFA) shall be available for users who want extra security. • The system shall follow OWASP security standards to prevent common attacks.
Usability	• A new user shall be able to complete registration and add their first transaction within 5 minutes. • The time taken by a user to create a new budget shall not exceed 3 minutes. • The error rate for adding transactions shall be less than 5%. • The interface shall be intuitive enough that 90% of users can navigate without training. • The app shall support both English and Arabic languages.
Reliability	• The system shall have 99.9% accuracy in all



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

	financial calculations. • No data loss shall occur during crashes or unexpected shutdowns. • The system shall automatically backup user data every 24 hours.
Portability and compatibility	The mobile app shall support Android versions 8.0 to 14.0. • The mobile app shall support iOS versions 13.0 to 18.0. • The web application shall work on Chrome (versions 80+), Firefox (versions 85+), and Safari (versions 14+). • The app shall work on both phones and tablets with responsive design.
Maintainability	The system shall be modular so that updating one feature doesn't affect others. • Code documentation shall be provided for all major functions. • Critical bugs shall be fixable within 48 hours.

System Models

Use Case Model



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Project: Personal Budgeting Software

Software Requirements Specifications

Actors Definition & Roles

1. User (Registered)

A registered individual who uses the Personal Budgeting System to manage their personal finances. The User is the primary actor in the system and interacts with the majority of its features. Their role includes logging income and expenses, creating and monitoring budgets, setting financial goals, tracking savings progress, viewing financial reports, and linking external bank accounts or e-wallets to automate transaction tracking.

2. Bank / E-Wallet (External System)

An external financial institution or digital payment platform (e.g., a bank, mobile wallet, or credit card provider) that integrates with the system via API or sync mechanisms. Its role is to supply the system with real-time or periodic transaction data, enabling automatic tracking of the user's financial activity without manual entry.

3. Admin

A privileged internal user responsible for maintaining and operating the system. The Admin does not manage personal budgets but oversees the platform as a whole. Their role includes managing user accounts (creating, suspending, or deleting them), configuring the default income and expense categories available to all users, and generating system-wide reports to monitor platform usage and performance.



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

Enriched User Stories

- **User Story #1**

User Story ID	US #1
User Story Name	User Sign-Up
Actors	New User (Player/Player equivalent)
Description	As a new user, I'd like to be able to sign up for a new account, .So that I can access the application's features
Per condition	The user is on the welcome screen and does not have an account.
Post condition	A new user account is created, and the user is logged in.
Acceptance Criteria	Given I am a new user on the sign-up page, When I fill in all required fields with valid information and click the "Create Account" button, Then the system creates my account and logs me into the application.

- **Scenarios**

Normal Scenario

Actor Action	System Response
1- User clicks on "Sign Up".	
	2- System displays the sign-up page with fields for name, email, and password.
3- User fills in the information and clicks "Create Account".	
	4- System verifies the provided information (e.g., email format, password strength).
	5- System saves the user information in the database.
	6- System automatically logs the user in



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

	and redirects them to the main dashboard.

Exceptional Scenario

Actor Action	System Response
1- User clicks on "Sign Up".	
	2- System displays the sign-up page.
3- User fills in the information (e.g., with an invalid email format) and clicks "Create Account".	
	4- System verifies the information and finds it invalid. 5- System displays an error message: "Please enter valid information." next to the incorrect field.

- **Screen Design**



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Project: Personal Budgeting Software

Software Requirements Specifications

- **Data Dictionary:**

Element Label	Type/Length	Data Validation / Business Rule
Full name field	Text <100 characters	Structured text. Must not be empty
Email field	Text <100 characters	Structured Text. Must be a valid email format (e.g., user@example.com). Must be unique in the system.
Password field	Alphanumeric & special chars < 100 Characters	Must be at least 8 characters long and include an uppercase letter, a number, and a special character.
Confirm Password field	Alphanumeric & special chars < 100 Characters	Must exactly match the "Password" field.



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

- **User Story #2**

User Story ID	US #2
User Story Name	Log in
Actors	Registered User (Player/Playground Owner equivalent)
Description	As a registered user, I'd like to be able to log in to my account, So that I can access my personal financial data and use the application's features
Per condition	The user is on the welcome screen and has an existing account.
Post condition	The user is successfully authenticated and granted access to the application.
Acceptance Criteria	Given I am a registered user on the log-in page, When I fill in the "Email" and "Password" fields with my correct credentials and click the "Log In" button, Then the system signs me in and takes me to my dashboard.

- **Scenarios**

Normal Scenario

Actor Action	System Response
1- User clicks on "Sign Up".	
	2- System displays the log-in page with boxes to input email and password.
3- User fills in the boxes with their credentials and clicks "Log In".	
	4- System verifies the user information against the database.
	5- System grants the user access to the main application dashboard.



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

Exceptional Scenario

Actor Action	System Response
1- User clicks on "log in".	
	2- System displays the log in page.
3- User fills in the boxes with incorrect credentials and clicks "Log In".	
	4- System verifies the information and finds it invalid. 5- System displays an error message: "Invalid email or password. Please try again."

• Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Email field	Text <100 characters	Structured Text. Must match an existing user's email in the database.
Password field	Alphanumeric & special chars < 100 Characters	Structured Text. Must match the password associated with the provided email.

• User Story #3

User Story ID	US #3
User Story Name	Add transaction
Actors	User (Player equivalent)
Description	As a user, I'd like to be able to add a new financial transaction, .So that I can keep a record of my income and expenses
Per condition	The user is logged in and is on the "Transactions" screen or the main dashboard.
Post condition	A new transaction is saved to the user's account and reflected



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Project: Personal Budgeting Software

Software Requirements Specifications

	in their balance, budgets, and reports.
Acceptance Criteria	Given I am a logged-in user, When I navigate to the "Add Transaction" screen, fill in the details (type, amount, category, etc.), and click "Save", Then the system saves the transaction and updates my financial overview.

- Scenarios**

Normal Scenario

Actor Action	System Response
1- User clicks the "+" (Add) button on the dashboard or transactions screen.	
	2- System displays the "Add Transaction" screen.
3- User selects "Expense" or "Income" using a toggle.	
4- User taps on the amount field and enters the amount using a keypad.	
5- User selects a category (e.g., Food, Transport) from a list.	
6- User optionally adds a description, date/time, or note.	
7- User clicks the "Save Transaction" button.	
	8- System validates the input (e.g., amount is a positive number). 9- System saves the transaction to the database. 10- System displays a success message and redirects the user back to the previous screen, with the new transaction visible in the list.



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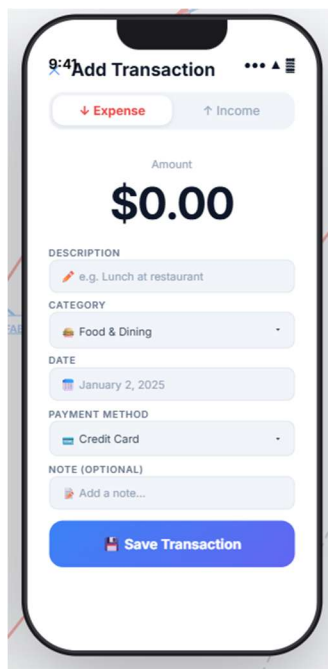
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Software Requirements Specifications

Exceptional Scenario

Actor Action	System Response
1- User navigates to the "Add Transaction" screen.	
	2- System displays the "Add Transaction" screen.
3- User fills in some details but forgets to enter an amount and clicks "Save Transaction".	
	4- System validates the input and finds the amount field empty. 5- System displays an error message: "Please enter an amount."

- Screen Design



- Data Dictionary:



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

Element Label	Type/Length	Data Validation / Business Rule
Transaction Type	Boolean (Income/Expense)	Determines if the amount is added to or subtracted from the balance.
Amount	Decimal (10, 2)	Must be a positive number greater than 0. Max 99999999.99.
Category	Text < 50 Characters	Selected from a predefined or user-created list. Cannot be empty.
Date & Time	Date/Time	Defaults to the current date/time, but can be changed. Format: DD/MM/YYYY, HH:MM.
Description	Text < 200 Characters	Optional free-form text.
Payment Method	Text < 50 Characters	Optional. Selected from a predefined list (e.g., Cash, Credit Card).

• User Story #4

User Story ID	US #4
User Story Name	Create/Edit Budget
Actors	User
Description	As a user, I'd like to be able to create and edit monthly budgets for different spending categories, So that I can control my spending and ensure I don't overspend
Per condition	The user is logged in.
Post condition	A new budget is created, or an existing budget is updated. The system will now track spending against this budget.
Acceptance Criteria	Given I am a logged-in user on the Budgets screen, When I choose to create a new budget or edit an existing one, fill in the details (category, amount, period), and save, Then the system creates/updates the budget and begins tracking my expenses against it.



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

- Scenarios

Normal Scenario

Actor Action	System Response
1- User navigates to the "Budgets" screen and clicks the "+ New" button (or taps on an existing budget to edit).	
	2- System displays the "Create Budget" (or "Edit Budget") screen.
3- User selects a category (e.g., "Food & Dining") from a dropdown.	
4- User enters the budget amount (e.g., \$300).	
5- User optionally sets the budget period (defaults to current month) and alert preferences.	
6- User clicks the "Create Budget" button.	
	7- System validates the input (e.g., amount is a positive number).
	8- System saves the budget to the database. 9- System redirects the user to the main "Budgets" screen, where the new/updated budget is now visible.

Exceptional Scenario

Actor Action	System Response
1- User goes through the steps to create a new budget.	
2- User selects a category for which a budget already exists for the same month.	
3- User enters the amount and clicks "Create Budget".	
	4- System checks for an existing budget and finds a conflict. 5- System displays an error message: "A budget for this category already exists for



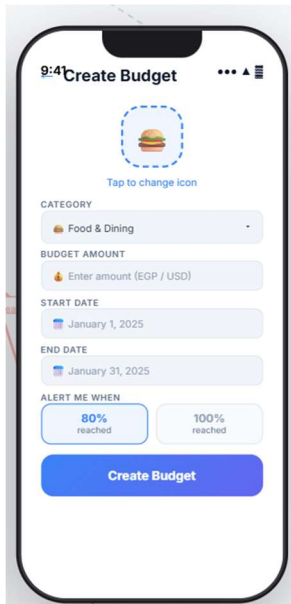
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Project: Personal Budgeting Software

Software Requirements Specifications

this month. Please edit the existing budget instead."

- **Screen Design**



- **Data Dictionary:**

Element Label	Type/Length	Data Validation / Business Rule
Category	Text < 50 Characters	Selected from a list of spending categories. Must be unique per user per month.
Budget Amount	Decimal (10, 2)	Must be a positive number greater than 0.
Start Date	Date	The first day of the budget period. Defaults to the 1st of the current month.
End Date	Date	The last day of the budget period. Defaults to the last day of the current month.
Alert Threshold	Integer	A percentage (e.g., 75, 100) at which the user wants to be notified.



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Project: Personal Budgeting Software

Software Requirements Specifications

- User Story #5

User Story ID	US #5
User Story Name	Budget Over-Limit Alert
Actors	User, System
Description	As a user, I'd like the system to alert me when I am close to or have exceeded my budget for a category, .So that I can adjust my spending habits in real-time
Per condition	The user has active budgets. The user has added a transaction that pushes a budget category to or past its limit.
Post condition	The system generates and displays an alert to the user.
Acceptance Criteria	Given I have a budget for a category, When I add an expense that causes my spending in that category to reach 100% or more of my budget, Then the system immediately shows me a visual alert (e.g., a red progress bar) and/or a notification.

- Scenarios

Normal Scenario

Actor Action	System Response
1- User adds a new expense transaction for a category with an active budget (e.g., adds a \$50 grocery expense to a \$200 grocery budget).	
	2- System calculates the new total spent for that budget category.
	3- System finds that the total spent (\$195) is now at 97.5% of the budget.
	4- System updates the budget's progress bar to show it is almost full and changes its color to orange or red.
	5- System creates a notification for the



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Project: Personal Budgeting Software

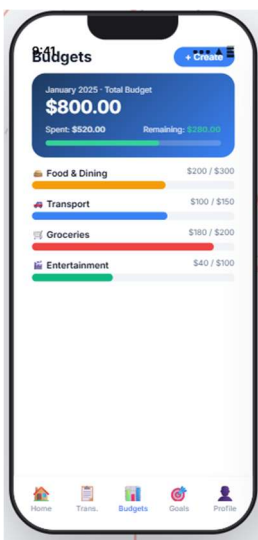
Software Requirements Specifications

user: "Budget Alert — Groceries: You've used 97.5% of your Groceries budget."

Exceptional Scenario

Actor Action	System Response
1- User adds a new expense transaction for a category with an active budget (e.g., adds a \$50 expense to a \$200 budget, where \$180 was already spent).	
	2- System calculates the new total spent (\$230) and finds it has exceeded the budget by \$30.
	3- System immediately updates the budget's progress bar to 115% and turns it red.
	4- System creates a prominent notification: "Budget Exceeded — Groceries! You've exceeded your \$200 budget by \$30." 5- This alert also appears on the main dashboard for immediate visibility.

- Screen Design





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Project: Personal Budgeting Software

Software Requirements Specifications

- **Data Dictionary:**

Element Label	Type/Length	Data Validation / Business Rule
Budget ID	Integer	Unique identifier for the budget.
Spent Amount	Calculated Decimal	Sum of all expense transactions for the category within the budget period.
Budget Status	Calculated Text	Derived from (Spent Amount / Budget Amount). Values: "On Track", "Near Limit", "Exceeded".

- **User Story #6**

User Story ID	US #6
User Story Name	Set and Track Financial Goals
Actors	User
Description	As a user, I'd like to be able to set financial goals (e.g., saving for a vacation) and track my progress, So that I stay motivated and see how my savings are adding up over time
Per condition	The user is logged in.
Post condition	A new financial goal is created and tracked.
Acceptance Criteria	Given I am a logged-in user on the Goals screen, When I click "Add Goal", fill in the details (name, target amount, deadline) and save, Then the system creates my goal and displays it with a progress bar on the Goals screen.

- **Scenarios**

Normal Scenario



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Project: Personal Budgeting Software

Software Requirements Specifications

Actor Action	System Response
1- User navigates to the "Goals" screen and clicks the "+ Add" button.	
	2- System displays a screen to create a new goal.
3- User enters a goal name (e.g., "Summer Vacation"), a target amount (e.g., \$5,000), and a deadline (e.g., Jun 30, 2025).	
4- User optionally sets an initial saved amount.	
5- User clicks "Create Goal".	
	6- System validates the input.
	7- System saves the goal and calculates the monthly savings needed.
	8- System redirects the user to the main Goals screen, where the new goal is now visible in the list with a progress bar.
9- (Later) User adds money to the goal by linking it to a transaction or manually adding a contribution.	10- System updates the goal's progress bar and the saved amount.

Exceptional Scenario

No exceptional scenario for this user story.

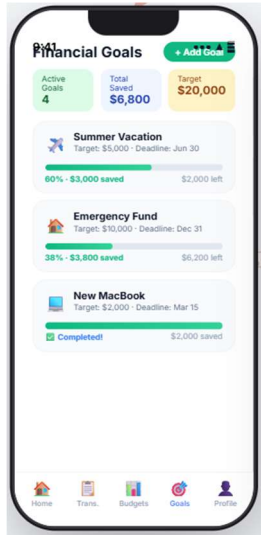
- **Screen Design**



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications



• Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Goal Name	Text < 100 Characters	Structured Text. Cannot be empty.
Target Amount	Decimal (10, 2)	Must be a positive number greater than 0.
Current Amount	Decimal (10, 2)	Must be a positive number, less than or equal to the Target Amount. Defaults to 0.
Deadline	Date	Must be a future date.
Status	Calculated Text	Derived from (Current Amount / Target Amount). Values: "In Progress", "Completed".

• User Story #7

User Story ID	US #7
User Story Name	View Reports and Analytics
Actors	User
Description	As a user,



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

	I'd like to see visual reports and charts of my income, expenses, and spending patterns, So that I can better understand my financial behavior and .identify areas for improvement
Per condition	The user is logged in and has recorded several transactions.
Post condition	The system generates and displays a financial report for the selected period.
Acceptance Criteria	Given I am a logged-in user on the Reports screen, When I select a date range (e.g., "This Month"), Then the system displays a pie chart of my expenses by category and a bar chart comparing my income and expenses.

- Scenarios**

Normal Scenario

Actor Action	System Response
1- User navigates to the "Reports" screen.	
	2- System displays a screen with an option to select a date range (defaulting to the current month).
	3- System fetches all transactions for that period.
	4- System generates and displays a pie chart showing the percentage breakdown of expenses by category.
	5- System generates and displays a bar chart comparing total income vs. total expenses for each week/month in the period.
	6- System displays a key insight message based on the data (e.g., "Food spending is 15% above average.").



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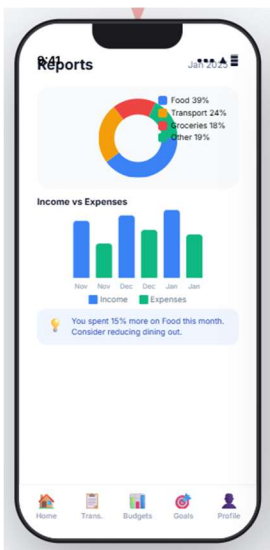
Project: Personal Budgeting Software

Software Requirements Specifications

Exceptional Scenario

Actor Action	System Response
1- User navigates to the "Reports" screen.	
	2- System displays the date range selector.
	3- User selects a date range for which there are no transactions.
	4- System displays a message: "No transaction data available for this period." The chart areas remain empty or show a placeholder graphic.

• Screen Design



• Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Report Period	Date Range	The start and end date for which the report is generated.
Category Totals	Calculated Decimal	Sum of all expense transactions for each category within the report period.
Income vs Expense Data	Calculated Data Set	Sum of all income and sum of all expense transactions, grouped by



CS251: Phase 1 –Investors

Project: Personal Budgeting Software

Software Requirements Specifications

Element Label	Type/Length	Data Validation / Business Rule
		day/week/month.

- User Story #8**

User Story ID	US #8
User Story Name	Edit Profile and Settings
Actors	User
Description	As a user, I'd like to be able to view and edit my profile information and application settings, So that my experience is personalized and my information is up-to-date
Per condition	The user is logged in.
Post condition	The user's profile or application settings are updated.
Acceptance Criteria	Given I am a logged-in user on the Profile screen, When I change any of my settings (e.g., name, currency, notification preferences) and save, Then the system updates my profile and applies the new settings throughout the app.

- Scenarios**

Normal Scenario

Actor Action	System Response
1- User navigates to the "Profile" screen.	
	2- System displays the user's profile picture, name, email, and account statistics (e.g., number of transactions, budgets).
	3- System displays a list of settings options



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Project: Personal Budgeting Software

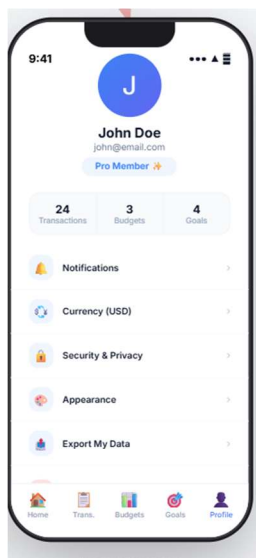
Software Requirements Specifications

	(e.g., Notifications, Currency, Language, Appearance).
4- User taps on "Currency".	
	5- System displays a list of available currencies.
6- User selects "EUR" instead of "USD".	
	7- System updates the user's preference in the database.
8- User taps the back button.	9- System returns to the Profile screen, and all future amounts will now be displayed in Euros.

Exceptional Scenario

No exceptional scenario for this user story.

- **Screen Design**





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Project: Personal Budgeting Software

Software Requirements Specifications

- Data Dictionary:**

Element Label	Type/Length	Data Validation / Business Rule
Display Name	Text < 100 Characters	Structured Text.
Currency	Text < 3 Characters	Selected from a list of supported currencies (e.g., USD, EGP, EUR). Affects how all monetary values are displayed.
Language	Text < 10 Characters	Selected from a list of supported languages (e.g., English, Arabic).
Notification Settings	Boolean	Multiple settings to turn on/off different types of notifications (e.g., budget alerts, goal reminders).

- User Story #9**

User Story ID	US #9
User Story Name	Filter and View Transaction History
Actors	User
Description	As a user, I'd like to be able to view and filter my complete history of transactions, So that I can review past spending, find a specific transaction, or analyze my activity for a certain category
Per condition	The user is logged in and has recorded multiple transactions.
Post condition	The system displays a filtered or sorted list of transactions.
Acceptance Criteria	Given I am a logged-in user on the Transactions screen, When I select a filter (e.g., by category "Food"), Then the system displays only the transactions that match that filter.

- Scenarios**



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Software Requirements Specifications

Normal Scenario

Actor Action	System Response
1- User navigates to the "Transactions" screen.	
	2- System displays a list of all transactions, grouped by date (e.g., "Today", "Yesterday"). It also shows summary cards for total income, expenses, and balance for the period.
3- User taps on the "Filter" icon or a category chip (e.g., "Food").	
	4- System re-queries the database and updates the list to show only transactions belonging to the "Food" category.
5- User removes the filter.	
	6- System displays the full, unfiltered list of transactions again.

Exceptional Scenario

Actor Action	System Response
1- User navigates to the "Transactions" screen and applies a filter (e.g., for a very specific category or date range).	
	2- System searches for transactions matching the filter.
	3- System finds no matching transactions.
	4- System displays a message in the transaction list area: "No transactions found for this filter."

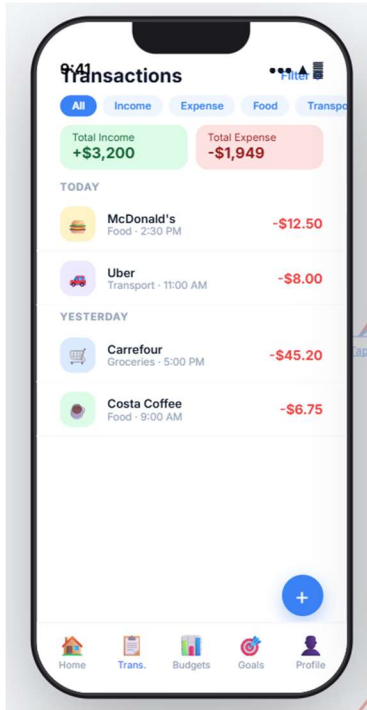
- **Screen Design**



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Software Requirements Specifications



- **Data Dictionary:**

Element Label	Type/Length	Data Validation / Business Rule
Filter Criteria	Text/Date Range	The parameter used to filter the transaction list (e.g., category name, date range).
Transaction List	Dynamic List	A list of transaction objects that match the current filter criteria.

- **User Story #10**

User Story ID	US #10
User Story Name	Dashboard Overview



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Software Requirements Specifications

Actors	User
Description	As a user, I'd like to see a dashboard when I open the app that gives me a quick snapshot of my financial health, So that I can immediately see my balance, recent activity, and budget status without having to navigate through multiple screens
Per condition	The user is logged in.
Post condition	The system displays the main dashboard with key financial information.
Acceptance Criteria	Given I am a logged-in user, When I open the application or navigate to the Home screen, Then the system displays my current total balance, a summary of recent transactions, and a preview of my budgets.

- Scenarios**

Normal Scenario

Actor Action	System Response
1- User logs in and is directed to the home screen.	
	2- System fetches the user's latest data from the database.
	3- System displays a "Total Balance" card with the current balance, along with total income and expenses for the month.
	4- System displays small statistic cards for "Transactions", "Budgets", and "Goals".
	5- System displays a short list of the most recent transactions.
	6- System displays a warning if any budget is near or over its limit.



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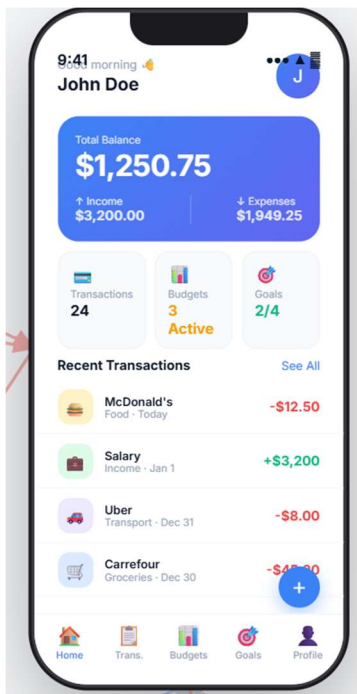
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Software Requirements Specifications

Exceptional Scenario

No exceptional scenario for this user story.

- **Screen Design**



- **Data Dictionary:**

Element Label	Type/Length	Data Validation / Business Rule
Total Balance	Calculated Decimal	Total Income - Total Expenses (overall).
Monthly Income	Calculated Decimal	Sum of all income transactions for the current month.
Monthly Expenses	Calculated Decimal	Sum of all expense transactions for the current month.
Recent Transactions	Dynamic List	The 3-5 most recent transactions from the database.



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Software Requirements Specifications

- User Story #11

User Story ID	US #11
User Story Name	Notifications Center
Actors	User, System
Description	As a user, I'd like to have a central place to view all my notifications, So that I don't miss important alerts about my budgets, goals, or .account
Per condition	The user is logged in and has received notifications.
Post condition	The system displays a list of all past and present notifications.
Acceptance Criteria	Given I am a logged-in user, When I navigate to the Notifications screen, Then the system displays a list of all my notifications, clearly marking which are unread.

- Scenarios

Normal Scenario

Actor Action	System Response
1- User taps on a notification bell icon or a "View All" link from an alert banner.	
	2- System displays the Notifications Center.
	3- System shows a list of all notifications, with the most recent at the top.
	4- Unread notifications are visually distinct (e.g., a different background color or a blue dot).
5- User taps on a notification.	
	6- System marks the notification as read.



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Software Requirements Specifications

	7- System may navigate the user to the relevant screen (e.g., the budget that triggered the alert).
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Exceptional Scenario

Actor Action	System Response
1- User navigates to the Notifications Center.	
	2- System checks for notifications and finds none.
	3- System displays a message: "No new notifications." and maybe a list of older, read notifications.

- Data Dictionary:**

Element Label	Type/Length	Data Validation / Business Rule
Notification ID	Integer	Unique identifier for the notification.
User ID	Integer	The ID of the user the notification belongs to.
Type	Text < 50 Characters	The category of notification (e.g., "BUDGET_ALERT", "GOAL_COMPLETE").
Message	Text < 255 Characters	The content of the notification.
Is Read	Boolean	A flag indicating whether the user has viewed the notification.
Timestamp	DateTime	The date and time the notification was generated.



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Software Requirements Specifications

- User Story #12

User Story ID	US #12
User Story Name	Export Financial Data
Actors	User
Description	As a user, I'd like to be able to export my financial data, So that I can have a backup copy, analyze it in a spreadsheet, or share it with a financial advisor
Per condition	The user is logged in and has recorded transactions.
Post condition	A file containing the user's financial data is generated and downloaded.
Acceptance Criteria	Given I am a logged-in user on the Export Data screen, When I select a format (e.g., CSV) and a date range, and click "Generate & Download", Then the system generates a file with all my transactions for that period and initiates a download.

- Scenarios

Normal Scenario

Actor Action	System Response
1- User navigates to the "Export Data" screen from the Profile or Settings menu.	
	2- System displays options for "Format" (PDF, CSV, Excel) and "Date Range".
3- User selects a format (e.g., CSV) and a date range (e.g., "This Month").	
4- User optionally selects what to include (Transactions, Budget Summary, Goals).	
5- User clicks the "Generate & Download" button.	
	6- System validates the request.



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Software Requirements Specifications

	7- System queries the database for all relevant data.
	8- System formats the data into the selected file type.
	9- System initiates a download of the file to the user's device.
	10- System displays a success message: "Your file has been generated and downloaded."

Exceptional Scenario

No exceptional scenario for this user story.

• Data Dictionary:

Element Label	Type/Length	Data Validation / Business Rule
Export Format	Text < 10 Characters	Selected by user. Must be one of: "PDF", "CSV", "EXCEL".
Date Range	Date Range	The start and end date for the exported data.
Data Inclusions	Boolean Flags	A set of flags indicating which data modules (Transactions, Budgets, Goals) to include in the export.
Generated File	File	A file object created by the system in the requested format.

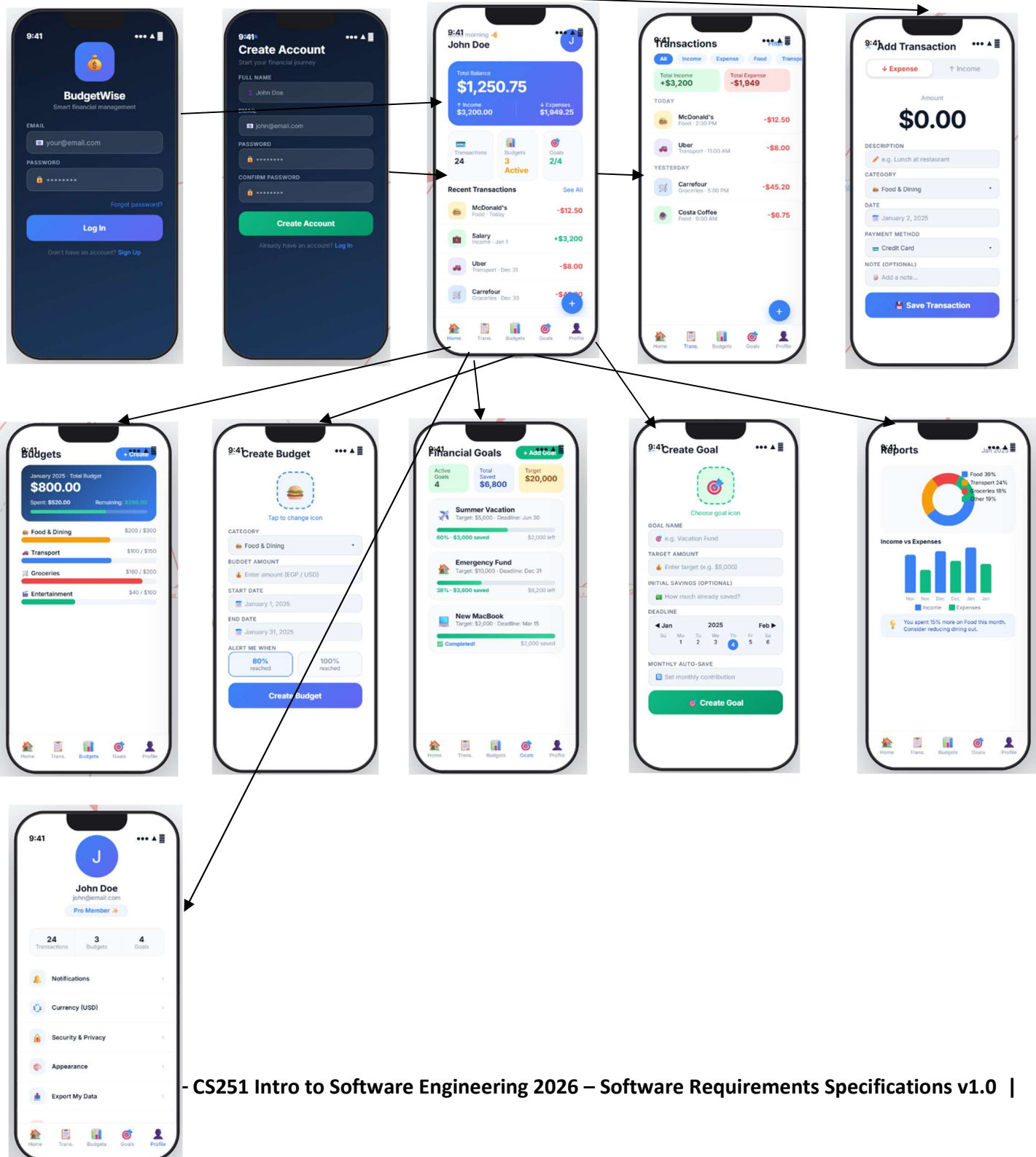


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Software Requirements Specifications

System Navigation Map





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Project: Personal Budgeting Software

Software Requirements Specifications

Tools

- MockFlow, Claude

Ownership Report

Student	Items he created
Renad Tharwat yehia	Part of user stories- Nonfunctional Requirements- Nav Map- Document
Manal Abdelazeem Abdelrahman	Functional Requirements- Nonfunctional Requirements-Documents
Noura Mohammed Mokhtar	Part of use case model- Definitions and Abbreviations- Nav Map
Menna Tarek Saad	Part of use case model- Part of user stories